EXTENDING BOUNDARY COMPOSITION TO THE COMPLETED NEVANLINNA CLASS

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> It remains an open question whether Lemma 4.4 can be extended to the full Nevanlinna class $\mathcal{N}$.
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> Allu and Mondal, *Composition operators and Carleson embeddings for the Nevanlinna class of Dirichlet series*, arXiv:2609.17133, Remark 4.5, pp. 27–28.

Here $\mathbb{C}_0 = \{\operatorname{Re} s > 0\}$, and Lemma 4.4 is the boundary push-forward identity $\|f \circ \Phi\|_0 = \int_{\mathbb{T}^\infty} \log(1 + |f(\Phi^*(\chi))|)\, \mathrm{d}m(\chi)$ for $f \in \mathcal{N}_u$, where Φ^* is the almost-everywhere boundary value of its Bohr lift and $\|\cdot\|_0$ is the limiting logarithmic mean. The completed class $\mathcal{N}$ is the metric completion for $d_0(f, g) = \|f - g\|_0$.

Answered: no. Evidence: Sketch.

No. For a symbol satisfying the Gordon–Hedenmalm hypotheses, a natural scalar composition of a completed-class input need not belong to the completion.

Proof. Choose a smooth $g \geq 0$ on $\mathbb{T}$ with mean $3/2$ that vanishes on an open arc, and put

$$\psi(z) = \frac{1}{2} + \int_{\mathbb{T}} \frac{\xi + z}{\xi - z} g(\xi)\, \mathrm{d}m(\xi), \qquad \Phi(s) = \psi(2^{-s}).$$

Smoothness gives a bounded boundary extension; $\operatorname{Re}\psi \geq 1/2$ on the closed disc, and $\operatorname{Re}\psi > 1$ for $|z| < 1/2$. Hence Φ has characteristic zero, maps $\mathbb{C}_1$ into $\mathbb{C}_1$, and has relatively compact range in $\mathbb{C}_0$.

Let $A = 64$, $a_k = A^k$, $N_k = \lceil \exp(\exp(A^{8k})) \rceil$, $L_k = \log N_k$, and

$$F(s) = \sum_{k \geq 1} \frac{a_k}{\sqrt{N_k}} P_{N_k}(s), \qquad P_N(s) = \sum_{n \leq N} n^{-s}.$$

Harper's *Moments of random multiplicative functions, I: Low moments, better than square-root cancellation, and critical multiplicative chaos*, Theorem 1, doi:10.1017/fmp.2019.7, gives $\|P_{N_k}\|_{\mathscr{H}^1} \leq C\sqrt{N_k}(\log\log N_k)^{-1/4}$, so the block norms are summable and $F \in \mathscr{H}^1$. Its ordinary series converges absolutely on $\operatorname{Re} s > 1/2$. Comparison of sums and integrals gives $P_N(s) = N^{1-s}/(1-s) + O_T(1)$ uniformly for $1/2 \leq \operatorname{Re} s \leq 3/4$ and $|\operatorname{Im} s| \leq T$. The kth main term dominates on the band $\alpha/L_k \leq \varepsilon \leq 1/L_k$: earlier terms sum to at most $a_k/63$, later terms to less than $a_k/100$, and the error is bounded. Consequently, for each fixed $\alpha, T > 0$,

$$|F(1/2 + \varepsilon + it)| \geq b_T a_k \quad (\alpha/L_k \leq \varepsilon \leq 1/L_k,\ |t| \leq T)$$

for all large k.

Suppose some completed-class element h had canonical scalar evaluations agreeing with $F(\Phi(s))$ on a right half-plane. Its holomorphic Bohr lift B_h is defined on $\ell^1 \cap \mathbb{D}^\infty$. The slice $b(z) = B_h(z, 0, \ldots)$ lies in the classical disc Nevanlinna class: the completed gauge bounds the weighted radial means of $\log(1 + |B_h|)$, and successive submean inequalities in coordinates after the first, followed by the uniformly small ℓ^1 tail, bound the radial means of $\log(1 + |b|)$.

For fixed sufficiently large σ, the prime vertical orbit is dense in the torus of radii $(p_j^{-\sigma})_j$, since the prime logarithms are rationally independent. Continuity of the Bohr lift therefore extends the assumed agreement from that orbit to all phases. Integrating over coordinates after the first and using the holomorphic mean-value identity gives $b(2^{-\sigma}\chi_1) = G(2^{-\sigma}\chi_1)$, where $G = F \circ \psi$. Thus $b = G$ on $\mathbb{D}$ by the identity theorem.

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On a closed positive-measure arc I inside the zero arc of g, the Poisson formula gives $\operatorname{Re}\psi(r\zeta) - 1/2 \asymp 1 - r$, uniformly for $\zeta \in I$, while $|\operatorname{Im}\psi(r\zeta)|$ is bounded. Taking $r_k = 1 - 1/(CL_k)$ puts the real margin in the band above, so $|G(r_k\zeta)| \geq b_T A^k$ on I. Hence the radial means of $\log(1 + |G|)$ diverge, contradicting $G = b \in N(\mathbb{D})$. This excludes a natural completed-class composition output and therefore any continuous extension that agrees with composition. The proof uses no surjectivity assumption for the completed Bohr lift. □

AI assistance was used in adapting and checking the argument and the exposition.